

Making Best Use of Performance Measures and Information

Ellen J. Dumond

California State University, Fullerton, California, USA

Introduction

Managing any function or individual can be difficult and managing a boundary-spanning function is even more challenging. One such boundary-spanning organization is procurement, which must link a firm's internal operations with external sources of supply. Because of the breadth of activities and responsibilities and the magnitude of monetary resources they control, managing the procurement function is especially challenging. Its expenditure generally accounts for as much as 63 per cent of the total revenues of a corporation through the purchase of raw materials, components and subassemblies, as well as capital equipment. These materials affect all elements of the firm. Every defect in the material increases costs either directly by delaying production or indirectly by reducing the value of the product to the customer. Every late delivery to the plant increases costs. To be effective, this function must interact with external suppliers as well as numerous internal organizations, such as engineering, production and marketing. In the past, when raw materials were relatively cheap and simple, when manufacturing processes were straightforward, when engineering changes were few, when masses of stock was considered an asset, when the term "quality" was reserved for engineers, the procurement function could survive as an island seeking to cut costs. Now, that approach is not feasible. The complexity of raw materials and manufacturing processes, demanding customers, and substantial competition make it essential that "quality" be sought throughout every element of the plant. The movement to synchronized manufacturing and an elimination of buffer stock mandate improved processes – including the procurement of the right material, at the right place, at the right time.

While some firms have recognized the need to change, others have not. This may explain why, across firms, the procurement function is often managed very differently. As it is integrated into the strategy of the firm, procurement individuals are being asked to perform new roles and responsibilities. Among other things, they are expected to become team players, individuals capable of making strategic, proactive decisions for the firm. The manner of its management can affect both the decisions and comfort level of the procurement professionals and, ultimately, the performance of the firm.

The objective of this research is to examine the effects of different performance measurement systems (PMSs) and greater access to supporting information about internal operations and external events on both the decision-making and comfort level of boundary-spanning professionals. The article begins with a discussion of relevant research and continues with a discussion of the research method and experiment variables. Next, the research results are discussed. The final section presents the implications of these results for managers.

Relevant Literature

This research draws on the literature in three primary areas: performance measurement system, internal interaction and information about external events. Because each of these areas abounds with studies and discussions only selected literature is reported here.

Performance Measurement System

Performance measurement is a research topic that spans many disciplines. It involves the development of goals and related performance measures and the provision of feedback. The literature discusses the purposes of performance appraisal; presents various methods to measure performance; and examines numerous methods by which to establish evaluation criteria[1-3]. Within the procurement area itself, the literature abounds with descriptions of various performance measures and methods to determine them[4-7]. It is interesting to note, however, that while there is much written, there is little consensus regarding definitions, methods of measurement, or even what should be measured.

Because procurement serves many customers – the company itself, other internal functions, the procurement professionals and the suppliers – several goals are needed to cover all vital responsibilities. Areas in which procurement goals and performance measures are frequently established include department costs, productivity, price, incoming quality, supplier profile, inventory, incoming on-time, order cycle time and documentation[6,8]. Performance measures are established to support the achievement of goals and are provided with the intent to motivate, guide and improve an individual's decision making. These measures can be categorized into areas such as workload, quality, operations or price. Additionally, two common categories are those of efficiency and effectiveness. Efficiency is a rather narrow concept, focusing on the internal workings of the function, and is generally defined as the amount of resources used to produce a unit of output[6,9]. Consequently, procurement efficiency measures are generally time- or cost-based. There is more confusion as to the definition of effectiveness but there is consensus that it is a broader, more global measure than efficiency. For example, effectiveness has been defined in terms of the degree to which a function meets its goals; the ability of the function to acquire needed resources; the internal health or internal processes of the

function; or the degree to which the function meets the needs of its constituencies[10].

Because the purpose of measurement is to guide and improve performance, providing feedback to individuals with regard to where they stand on the performance measures is essential[11]. This can enhance performance by providing motivation or information about the correctness and adequacy of work behaviour[12-14], and can also provide workers with a sense of accomplishment, competence and control[15].

Interaction with Other Internal Functions

Several problems may occur in the implementation and use of performance measurement systems, including the lack of information sharing across departments[16]. Communication processes provide information to all parts of the organization, allowing for decision making and task accomplishment. Organizational communication generally flows in two directions: horizontally or vertically. Horizontal communication (often viewed as interaction) is the flow that occurs both within and between departments, the primary purpose being that of co-ordination. This type of communication becomes increasingly important as the complexity of the task and the demands for information increase[17]. It is likely that as we continue to move into the information age vertical communication will become less important while horizontal communication becomes more important[18].

One form of research in horizontal communication is in terms of “required interaction”, a situation in which interdependence between tasks is necessary to perform the tasks properly, and “optional interaction”, when the opportunity for interaction is provided but this interaction is not required for task completion[19,20]. Job-related or required interaction can affect an individual's task performance positively.

Additional research in the communication network literature links communication, information and performance. This research indicates that “isolation” or “participation” may affect individual performance positively[20-22]. It also provides support for the idea that “participative” individuals are more committed to the organization and generally more satisfied than are “isolated” individuals.

Information about External Events

The external environment, often referred to as the organizational environment, consists of all elements outside the boundary of the organization and includes elements of competition, resources, technology, economic conditions and other elements that may in some way be related to the performance of the organization. An organization needs information from its environment to make decisions and plan for the future. Organizations or individuals that face more uncertainty generally require greater amounts of information[23,24]; however, too much information can overload individuals[25]. Information distortion, as well as the preferred format of the communication, can also be a problem[23].

Within this research, there is no intent to overload individuals with information nor to provide inaccurate information; rather, a controlled amount of directly supportive information is provided.

Research Method

This research presents subjects with 14 problems requiring decisions. Before responding to them, the subjects are provided with an orientation scenario that describes how they will be measured and rewarded, and the environment into which they have been randomly placed. Subjects assume the role of a procurement manager.

The subject is then presented, via a computer, with a sequence of 14 procurement problem situations, each of which includes a short narrative describing a problem and four possible ways in which to respond to it. The computer manipulates each treatment to provide more (or less) information about internal and/or external events. The problem description portrays a situation in which the individual must make a decision between characteristics such as quality, cost, delivery reliability, short-term gain and long-term benefit. The situations are based on existing purchasing cases and incorporate incidents that would typically be faced by procurement managers[26,27].

After the subject makes a decision in response to a problem situation, appropriate performance feedback is provided. All feedback is quantitative, outcome feedback which relates to the particular performance measures. The feedback is not cumulative and relates only to the most recent situation and decision; it is unique to the performance measurement system (PMS) and the decision made. The subjects move through the 14 situations, receiving feedback where appropriate.

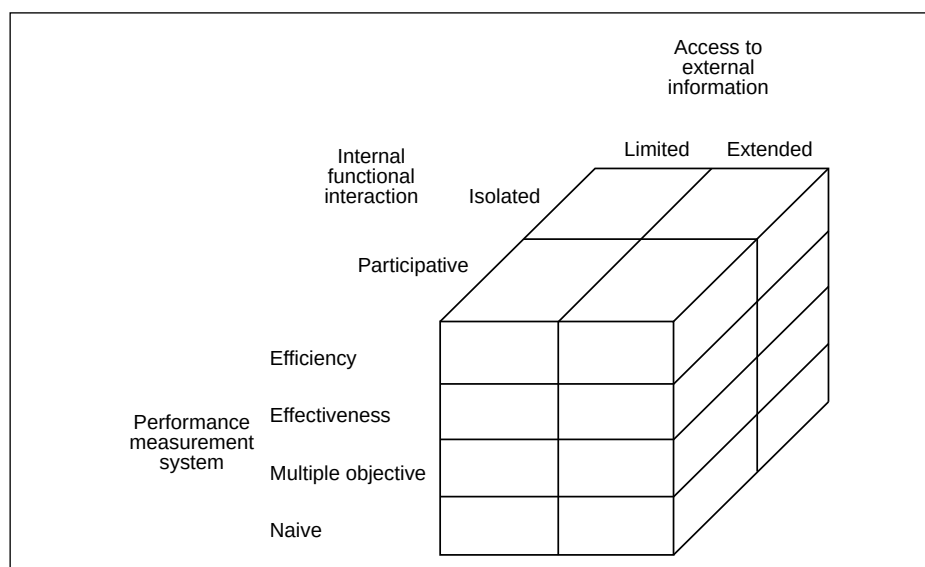
When finished with the problem situations, the decisions made by the subject are used to set parameters in a computer-based simulation of a factory. The results of the simulation establish reward levels and determine decision making effectiveness. The subjects also respond to questions about their attitudes and environment, using a Likert-like scale. Their responses are analysed to determine the subject's comfort level.

Manipulations

The research was conducted using a full-factorial experimental design, resulting in 16 ($4 \times 2 \times 2$) treatments. Fifteen subjects (upper division business school undergraduates) were in each treatment. The experimental treatments are depicted in Figure 1 and were operationalized in the following manner.

Performance measurement system. The performance measurement treatments reflect systems that can be found in today's procurement functions. The *efficiency* PMS reflects the traditional approach to procurement performance measurement, centring on minimizing costs and maximizing functional operating efficiency. An important section of the orientation scenario for the efficiency treatment reads: "Management has established two areas in which they evaluate your performance: the cost of purchases and the efficiency

Figure 1.
Experimental
Treatments



with which you perform your responsibilities. Your specific performance measures are cost of purchases (total annual material cost reductions and potential cost reductions) and efficiency (operating cost and order processing time)”.

The *effectiveness* PMS is more contemporary, highlighting the ability of procurement to contribute to the attainment of the firm’s goals. An important section of the orientation scenario for the effectiveness treatment reads: “Management has established three measures by which they evaluate your performance. Your specific measures are: contribution to profit, supplier relations and customer satisfaction. Of these three measures, although all of them are to be considered in your decisions, your total contribution to profit is the most important to management”.

A *multiple objective* PMS incorporates both efficiency and effectiveness. Subjects in this treatment were provided with an orientation scenario that identified all of the efficiency and effectiveness measures as being important to management and the basis for their performance measurement. The naive system is rather *laissez-faire*. Subjects placed in this treatment were provided with an orientation scenario with no performance measures – only to “do their best”. The performance measures for each PMS are illustrated in Table I. A general goal instructing subjects to “do their best” was used in all scenarios. Feedback was quantitative, outcome-based and provided only with regard to the performance measures.

Interaction with other internal functions. Internal functional interaction reflects the extent to which procurement communicates with the other internal functions. The heart of the concept is internal information flow; therefore, to operationalize the concept of functional interaction, the amount of information

Efficiency performance measurement system

Cost of purchases

- Total annual cost reductions
- Potential cost reductions

Efficiency

- Operating cost
- Order processing time

Effectiveness performance measurement system

Contribution to profit

- Actual contribution to profit
- Potential contribution to profit

Supplier relations

Customer satisfaction

Multiple objective performance measurement system

Both efficiency and effectiveness measures

Naïve performance measurement system

None

Table I.
Performance Measures
for Subjects

received by procurement from inside the firm is controlled. Subjects operating within an *isolated* environment receive little information about the firm's internal activities (only the basic scenario and problem descriptions) whereas subjects operating in a *participative* environment receive additional information about internal events for each problem situation. The subject is able to use this information in responding to the problem situation. This information may relate to product development, production equipment and processes, inventory status, production schedules, or other internal situations.

Information about external events. For this research, external information includes:

- other firms' experience with specific suppliers or materials;
- new materials or equipment in the marketplace;
- local, national, or international changes in taxes or regulations;
- long-term material availabilities; or
- future expected market conditions and trends.

Subjects operating in an environment with *extended* external information receive additional information for each problem situation while subjects operating in an environment with *limited* external information receive no additional information about external events.

Dependent Measures

The objective of the research is to investigate the effect of performance measurement systems, internal functional interaction and access to in-

formation about external events on an individual's decision-making performance and comfort level. Decision-making performance is based on the results achieved by the 14 decisions made by each subject. Each decision alternative has particular consequences associated with it. For example, a decision to acquire the lowest price component from an unreliable supplier might result directly in lower cost of materials, late deliveries or poor quality components and ultimately lead to reduced contribution to profit. In general, these effects (both direct and indirect) were captured through the simulation, in terms frequently associated with efficiency (e.g. order processing time, operating cost, cost of materials) and effectiveness (e.g. contribution to profit, customer satisfaction, supplier relations).

The decisions to the 14 problems were captured by the computer and served to set parameters for a simulation of a factory. The simulation, programmed in FORTRAN, simulates the production environment of a firm – generating demand for the end item, developing a master production schedule, placing orders for materials, shipping orders of the end item, and updating inventory records – for a period of 12 months. In the base case, the simulation performs in a “sterile” environment where deliveries of materials always occur as scheduled, the quality of the materials are such that no rework is required, the labour force is such that some, but little, overtime and idle time exists, price breaks are not encountered, etc. Consequently, the costs and profit serve as a base (or nominal case) against which to compare the subject's decision-making performance. To this model, the 14 procurement problem situations were added. The consequences (for example, quality problems or price breaks) of the subject's decisions were then incorporated into the simulation. The effects of the 14 decisions on the dependent variables are then captured and analysed.

The second category of dependent measures is the subject's comfort level about his/her performance and the particular environment in which he/she was operating. A post-experiment questionnaire addressed issues of understanding of performance measures, commitment, perceived performance, confidence, enjoyment, and usefulness of the feedback. The answers to these questions provide the basis for discussion of the subjects' comfort level.

Results

The results are discussed in terms of an individual's decision-making performance and comfort level. The analysis of the decision making is discussed first.

Decision Making

Decision-making performance refers to the impact of the decisions made by the individuals in the laboratory experiment. Measures of the effects of their decisions and their means are reported in Table II, which also presents the overall mean values for each measure and the theoretical range.

The maximum/minimum value for a single measure, also shown, could have been obtained if the subject chose the response on each of the 14 situations which maximized performance on that measure. However, it would have been unlikely for a subject to achieve the maximum or minimum for any two measures simultaneously because the decisions often had conflicting impacts.

As indicated in Table II, the PMS significantly influenced the decisions. Subjects in the naive PMS treatment made decisions that created a higher average order processing time than subjects in any of the other performance treatments – significantly higher than the effectiveness subjects (Scheffe 0.05).

	Means (Standard Deviation)					
	Average ^a order processing time (days)	Average operating cost	Average material cost reduction	Average contribution to profit	Average supplier relations	Average customer satisfaction
<i>Performance measurement system</i>						
Efficiency	14.70 (0.39)	2136 ⁺ (89)	4956*** (3709)	19616 (6792)	79.85 (1.04)	77.47 (1.65)
Effectiveness	14.58** (0.24)	2144 (98)	1506 (3204)	25931*** (6572)	80.92*** (1.08)	79.03*** (1.37)
Multiple objectives	14.64 (0.32)	2176 (113)	2235 (2475)	24507 (5495)	80.76 (1.03)	78.95 (1.11)
Naive	14.78 (0.32)	2173 (112)	2690 (2414)	25140 (6193)	80.56 (1.07)	78.82 (1.29)
<i>Functional interaction</i>						
Isolated	14.65 (0.38)	2166 (109)	3962*** (3216)	20744 (6453)	80.24 (1.15)	77.99 (1.53)
Participative	14.70 (0.28)	2148 (100)	1732 (2912)	26852*** (5550)	80.80*** (1.04)	79.15*** (1.25)
<i>Access to external information</i>						
Limited	14.68 (0.35)	2171 (101)	3095*** (3243)	21078 (6879)	80.09 (1.03)	77.89 (1.41)
Extended	14.67 (0.309)	2143* (107)	1789 (2925)	26519*** (5384)	80.95*** (1.06)	79.24*** (1.28)
Overall mean	14.7 (0.33)	2157 (105)	2847 (3264)	23798 (6749)	80.52 (1.13)	78.57 (1.51)
Maximum	16.14	2464	13536	37439	84.29	81.21
Minimum	12.86	1989	-5251	0	76.71	73.86
*** = significant at 0.001; ** = significant at 0.01; * = significant at 0.05; + = significant at 0.10 ^a Average is determined by summing the values over the 14 problem situations and dividing by 14						

Table II.
Decision-making
Performance Measures

This result indicates that subjects with a naive PMS tended to make more decisions that resulted in inefficient buying activities. Effectiveness subjects made decisions that significantly reduced average order processing time. Initially this result might be surprising because average order processing time is an efficiency measure, but some decisions made by the effectiveness PMS subjects to improve supplier relations (one of their personal performance measures) also reduced order processing times.

The PMS also impacts average operating costs ($p = 0.064$) but the Scheffe test at the 0.05 level was unable to distinguish between the groups. Efficiency and effectiveness subjects made decisions that resulted in lower than average operating costs. While this is intuitive for efficiency subjects because it is one of their personal performance measures, it is less intuitive for the effectiveness subjects. Recall, however, that effectiveness subjects make decisions to improve contribution to profit – and some of those decisions cause operating cost reductions.

The measure of material cost reduction reflects a reduction in material costs from the base case. As expected, performance on average material cost reductions was very strong with efficiency PMS subjects. They made decisions that reduced material costs significantly more than subjects in any other PMS (Scheffe 0.05). Effectiveness individuals obtained the lowest material cost reductions. Subjects with an efficiency PMS focused on achieving material cost reductions, as could be expected because it was their principal personal performance measure. It is also apparent that the multiple objective PMS subject weighed this measure against their other performance measures and achieved some reductions. Of some interest is that the naive PMS subject who was just told to “do his best” achieved more reductions than the multiple objective PMS subject, indicating a bias towards immediate cost savings.

With regard to the stated effectiveness measures (contribution to profit, supplier relations and customer satisfaction), the effectiveness PMS subjects made decisions that created significantly better results on all three. The contribution to profit is a broad measure, reflecting the extent to which the individual's decisions would conceptually contribute to the performance of the organization (based on the simulation results). Effectiveness PMS subjects, as would be expected, obtained significantly better results than efficiency PMS subjects (Scheffe 0.05). Perhaps of greater interest is that subjects in the multiple objective and naive treatments made decisions that yielded contributions to profit of the same magnitude as effectiveness PMS subjects. At least in this experiment, individuals who were provided with little direction or conflicting direction, made decisions that were more “effectiveness-like” than “efficiency-like”.

The supplier relations index reflects the theoretical relationship between the buying and selling firms. The subject was informed that the measure ranged from 1 to 100, with 100 being the best. Effectiveness and efficiency individuals had the highest and lowest values (respectively) for this measure. Efficiency subjects were significantly lower on this measure than were the other PMS

groups (Scheffe 0.05). Again, subjects in the multiple objective PMS and naive PMS treatments obtained average values above the mean and close to the performance of subjects in the effectiveness PMS treatment.

The customer satisfaction measure reflects the theoretical customer's satisfaction with the firm's product and is measured on a range of 1 to 100, with 100 being best. Effectiveness subjects made decisions that obtained the highest customer satisfaction while efficiency subjects were significantly lower on this measure than those in the other PMS treatments (Scheffe 0.05). Subjects in the multiple objective and naive treatments again obtained values above the mean and between those of efficiency and effectiveness subjects.

The second independent variable, interaction with other internal organizations, significantly affected four of the six decision-making performance measures. Individuals with expanded internal functional interaction (participative) made decisions that generated more contribution to profit, higher supplier relations and greater customer satisfaction than did isolated individuals. On the other hand, isolated individuals achieved significantly more material cost reductions than did participative individuals, although they contributed significantly less to profit.

Access to external information, the third independent variable, also significantly affected the subject's decisions, affecting five of the performance measures. Subjects with external information made decisions that created a lower operating cost, higher contribution to profit, better supplier relations and higher customer satisfaction than did those without the external information. However, subjects without access to external information consistently made decisions that generated significantly more material cost reductions than those with it.

There were few statistical interactions; the combination of PMS with internal functional interaction was significant on two of the measures (average operating cost and average material cost reductions), while the combination of internal functional interaction and access to external information was significant on one measure (average contribution to profit) at the 0.05 alpha level.

Internal functional interaction, when combined with all but the efficiency PMS, created lower operating costs. However, efficiency subjects who had interaction with other internal organizations generated higher operating costs than efficiency subjects without internal functional interaction ($p = 0.014$). It seems that the additional information about internal operations caused these subjects to make decisions (probably "profitable" ones) contrary to their personal performance measure. With regard to supplier relations, the nature of the interaction was quite clear. Subjects in all PMS treatments who had interaction with other internal organizations made decisions that developed significantly higher supplier relations ($p = 0.047$). It is likely that knowledge of internal operations allows procurement managers to avoid some of the conflict between supplier and buyer.

Access to information about internal events (participative functional interaction) or access to information about external events enhanced the tendency of the subject to make decisions that increased the contribution to profit. Moreover, access to both internal functional information and information about external events allowed the subjects to make decisions that provided significantly more contribution to profit ($p = 0.001$). This result occurred even though only half the subjects were told that contribution to profit was their personal performance measure.

The results of these analyses indicate that all three independent variables – PMS, internal functional interaction, and access to information about external events – influence the decisions made by an individual. Significant differences were obtained between the groups on all six decision-making performance measures.

Comfort Level

After addressing the problem situations, the subject completed a post-experiment questionnaire which asked for the subject's perceptions about his/her performance and the particular operating environment. More specifically, the questions asked whether the subject: understood his performance measurement system; was committed to the task; perceived that he performed well; was confident with the decisions he made; believed the feedback was useful; and enjoyed the environment. A seven-point Likert scale was used. The responses were analysed using analysis of variance with 15 observations per cell. Table III presents the mean values.

As indicated in Table III, of the three experimental variables, the performance measurement system had the most impact on comfort level. Those subjects in the effectiveness PMS treatment had the best understanding of how their performance was being evaluated (significantly more than either the efficiency or naive subjects – Scheffe 0.05); were more committed to improving their performance; were the most confident in their decisions (significantly more than multiple objective subjects – Scheffe 0.05); and enjoyed their environment the most. On the other hand, subjects in the naive PMS were significantly less committed to improving their performance, had significantly less understanding of how their performance was being evaluated, and enjoyed their environment less than effectiveness subjects (Scheffe 0.05). Yet, the naive subjects were among the highest in terms of confidence and perception of their performance (significantly higher than efficiency subjects – Scheffe 0.05). Subjects in the naive PMS also had the strongest feelings about whether feedback was, or would have been, useful. These subjects felt significantly more strongly that feedback would be useful than did others (Scheffe 0.05). It is also interesting to note that subjects in the efficiency treatment felt they had performed significantly worse than did those in any other performance treatment (Scheffe 0.05).

Access to information in general, whether about internal activities or external events, had little effect on the subject's perceptions. A very interesting

Factor	Means (Standard Deviation)					
	Understand personal evaluation	Commitment	Perceived performance	Confident	Feedback useful	Enjoy environment
<i>Performance measurement system</i>						
Efficiency	4.45 (1.20)	5.72 (1.01)	3.90 (1.07)	4.75 (1.26)	4.93 (1.34)	5.50 (1.17)
Effectiveness	5.30*** (1.08)	6.05*** (0.79)	4.58 (0.93)	4.93*** (0.82)	5.27 (0.88)	5.70** (1.09)
Multiple objectives	5.05 (1.06)	5.72 (0.74)	4.48 (1.02)	4.25 (1.24)	4.82 (1.16)	5.13 (1.36)
Naive	2.77 (1.70)	5.08 (1.56)	4.73*** (1.01)	4.90 (1.41)	6.57*** (0.67)	4.93 (1.27)
<i>Functional interaction</i>						
Isolated	4.61** (1.64)	5.68 (1.26)	4.34 (1.08)	4.60 (1.23)	5.38 (1.19)	5.18 (1.39)
Participative	4.18 (1.58)	5.61 (0.97)	4.51 (1.01)	4.82 (1.22)	5.41 (1.31)	5.45+ (1.10)
<i>Access to external information</i>						
Limited	4.38 (1.60)	5.67 (1.13)	4.34 (0.99)	4.63 (1.26)	5.53* (1.15)	5.39 (1.31)
Extended	4.40 (1.64)	5.62 (1.12)	4.51 (1.10)	4.79 (1.19)	5.26 (1.33)	5.24 (1.21)
<i>Overall mean</i>	4.39 (1.61)	5.64 (1.12)	4.43 (1.05)	4.71 (1.22)	5.40 (1.25)	5.32 (1.25)
*** = significant at 0.001; ** = significant at 0.01; * = significant at 0.05; + = significant at 0.10						

Table III.
Comfort Level Measures

result here is one that was *not* significant rather than significant. Previous communication research had supported the hypothesis that individuals having access to information about internal activities would be more committed to performance and to the organization but this research provided no additional support for that idea. Although in the “right” direction, the relationship was not significant. This finding questions the ability to increase an individual’s commitment through current efforts of teaming and empowerment – a key component of which is internal interaction. Subjects provided with information about internal events understood their personal performance measures significantly *less* than those who were not given information about internal activities. Apparently, subjects who were told that they were being personally

measured on local efficiency measures (e.g. operating costs, order processing time) were confused when they were provided with information about other internal activities. Additionally, subjects with limited information about external activities thought the feedback was (or would have been) more useful than those with greater access to that information. It is also worth noting that individuals having access to information about other internal activities were, on average, more confident in their decision making, believed they performed better, felt the feedback was useful, and enjoyed the environment more than those with limited information about them.

Discussion and Implications for Management

The primary purpose of this research was to investigate the effects of the performance measurement system, internal functional interaction and access to information about external events on an individual's decision making and comfort level. The results show that the PMS is most important in guiding an individual's performance and can have even greater effect when the right types of interaction and information are provided to support that PMS.

As indicated by the analysis of the decision-making performance, it is possible, through the PMS, to develop a consistent pattern of behaviour on the part of individuals making decisions. The PMS (a combination of goals, performance measures and feedback) guided the individual to make decisions towards a particular end; that is, the efficiency PMS subjects made decisions that reduced material and operating costs and the effectiveness PMS subjects made decisions that focused on improving the broader, firm-oriented measures of performance. The multiple objective and naive subjects, lacking a clear focus, were inconsistent in their decisions – creating, for the most part, mediocre performance. The provision of no direction or numerous measures that are not mutually reinforcing makes it more difficult to control or guide performance. The PMS affects not only an individual's decisions, but also the comfort level about that environment and his/her performance. The effectiveness PMS had the most positive effect in that these were the most confident, committed, and satisfied with both their performance and environment. Individuals in the naive PMS who were provided with no performance measures and no performance feedback were the most confused about their performance and the least committed to improving it – continuing to illustrate the importance of providing goals, performance measures, and feedback on those measures. Interestingly, however, these individuals had the highest perception of their performance – in some sense, they believed they were doing well when there was no PMS to be measured against. The lack of performance feedback provided no indication they were performing poorly; so, apparently, they established their own basis for evaluation. The provision of too many, or conflicting, performance measures created the opposite reaction. The multiple objective subjects had the least confidence in their decisions. They were also very dissatisfied with their performance. Written comments revealed that a ranking of the personal performance measures would have been useful.

With regard to the other two independent variables, information about internal activities or external events, the results support previous research: the existence of this information affects decision-making effectiveness. In general, the availability of more information (internal or external) caused the individual to make decisions to improve overall firm performance, rather than focus on narrower performance criteria like cost reduction. The more information made available, the stronger this tendency. Such information may temper an individual's tendency to focus on a particular measure at the expense of others. The availability of information (internal or external) has much more impact on the individual's decisions than on his/her comfort level.

It is important that management provide goals, performance measures, and feedback. Without them, there is no focus and decisions are erratic. On the other hand, the provision of too many measures or measures that are not mutually reinforcing can create confusion and erratic performance. Management needs to analyse the situation and the organization's goals and use performance measures that will encourage the desired behaviour. This is an important step when managing boundary spanning functions such as procurement which frequently require a wide range of measures. In some cases, these functions are provided with very few, if any, specific measures, relying on the "manager's good sense" to guide behaviour. Clearly, this research has shown the need to provide specific goals and measures. On the other hand, managers of these boundary-spanning functions are often given too many objectives that are not mutually reinforcing – that is, they are told to "take care of the needs of the production supervisor, the marketing manager, and make sure you cut costs". This situation creates frustration and unfocused decision making. The goals and performance measures should be as clear, consistent, and non-conflicting as possible so that individuals will understand them and confidently make decisions in accordance with them. This research has shown that individuals have a greater "comfort level" if they understand their performance measurement system. They feel more confident in their decision making and enjoy the environment more when they understand what is expected of them. They also want feedback about their performance relative to these measures.

Within the procurement and operations arena, the use of broader, process-oriented measures is growing. These measures, similar to the effectiveness PMS orientation used in this research, should generate better decision-making performance and more satisfied individuals. If the broader firm or process performance measures are to be used, managers should also encourage and, more importantly, enable their employees to obtain information about both the internal activities and external environment. Such information will enhance the individual's ability to meet the company's goals as well as their his/her personal performance measures. On the other hand, if narrow, departmental performance measures are used, the availability of such information is not critical.

Researchers must provide more answers for managers. The methodology used in this research is useful for evaluating other elements of performance

measurement systems and many issues have yet to be investigated. The development and investigation of goals in a total quality or just-in-time environment has not been performed and such a study is needed. Research about feedback in these environments is also needed. Outcome-based feedback was used in this study; while it can identify the need to change direction, it does not provide appropriate information as to "how to adjust". Efforts are needed to determine if a particular type of interaction (internal or external) would create more commitment or if it is totally dependent on the existence of broader, "effectiveness-like" performance measures. In short, this research has begun the investigation of management and organization variables within a boundary-spanning environment, but much more research is needed.

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